



**Department of Computer Science and
Engineering
National Institute of Technology Karnataka,
Surathkal – 575025**

Course Name : Compiler Design Lab

Course Code : CS304

Lab Number : Week 6 (S2)

Experiment Title : **Lexical Analysis and Parsing using LEX and YACC**

Question 1: Write a Lex and Yacc program that performs lexical analysis and parsing for basic loop constructs in a C-like language. Your program should:

- Tokenize keywords (for, while, int, float), identifiers, integer and floating-point numbers, arithmetic operators (+, -, *, /), relational operators (<, >, <=, >=, ==, !=), and delimiters ((,), {, }, ;).
- Print each token along with its lexeme and the corresponding line number. Report lexical errors for invalid tokens.
- Parse for and while loops, allowing nested loop constructs.
- Implement syntax error detection for the following cases:
 - Missing semicolon in a for loop header
 - Missing condition in a while loop
 - Missing closing brace (})
 - Unrecognized loop constructs (invalid keywords)
 - Unsupported update expressions in for loops



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Question 2: Write a Lex and Yacc program that performs lexical analysis and parsing for function definitions in a C-like language. Your program should:

- Tokenize keywords (for, while, int, float), identifiers, integer and floating-point numbers, arithmetic operators (+, -, *, /), relational operators (<, >, <=, >=, ==, !=), and delimiters ((,), {, }, ;).
- Print each token along with its lexeme and the corresponding line number. Report lexical errors for invalid tokens.
- Parse valid function definitions with typed parameters and return statements.
- Detect and report the following syntax errors:
 - Missing return statement in non-void functions
 - Missing parameter types in function headers
 - Missing semicolons in function bodies
 - Invalid nested function definitions
 - Conflicting types in multiple return statements

**Code and
Results
Submission
Deadline**

: Between 11:30 – 11:45 AM on the Lab Day
